

1 Using Quantity Units

General

You can manage quantities in the system using the following quantity units:

Base quantity unit

The base quantity unit is an objective description of the material used in an order. Using the base quantity unit, you can compare scrap of different operations, for example. It is therefore the header unit of the order.

Primary quantity unit

The primary quantity unit is the quantity unit used to record quantities. It can deviate from the base quantity unit.

The data collection is generally performed in the primary quantity unit. Every time data is posted, the quantity is converted into the secondary quantity unit or tertiary quantity unit, if defined accordingly. To convert the data, the system uses the conversion factors stored for the operation or a formula stored in the system.

When you transfer components for an operation, the input quantity refers to the production of an article in primary quantity unit.

Secondary quantity unit

The secondary quantity unit is an alternative quantity unit used with material on coils, for example.

Tertiary quantity unit

Further alternative quantity unit, like the secondary quantity unit.

Usage in the backlog of orders

Proceed as follows to convert two quantity units:

If conversion factors (numerator and denominator each > 0) are defined for the operation, these factors are used. Otherwise, the system searches a formula for these two quantity units in the formula management. If a formula is found, the system uses the formula to convert the quantity units. If no formula is found and the quantity units are identical, the system uses the quantity of the "from quantity unit" one-to-one as "to quantity unit". If no formula is found and the quantity units are not identical, the value of the "to quantity unit" is set to 0.

If the target quantities of an operation are converted, the system always converts all quantities. Here, the system uses the quantity (quantity unit) that the user has changed or that has been transferred via interface. If the quantities of different quantity units are changed, the system converts the quantities using the following priority – if conversion factors or a formula are available:

1. Primary quantity: conversion into secondary quantity and tertiary quantity
2. Secondary quantity: conversion into primary quantity and tertiary quantity
3. Tertiary quantity: conversion into primary quantity and secondary quantity

The base quantity unit is generally used for conversions between primary, secondary and tertiary quantities.



To specify the conversion factors (numerator, denominator) for the primary, secondary and tertiary quantity of an operation, always define the conversion INTO the base quantity unit.

$$\text{base quantity} = \text{primary quantity} * \frac{\text{numerator}_{\text{primary quantity}}}{\text{denominator}_{\text{primary quantity}}}$$

$$\text{base quantity} = \text{secondary quantity} * \frac{\text{numerator}_{\text{secondary quantity}}}{\text{denominator}_{\text{secondary quantity}}}$$

$$\text{base quantity} = \text{tertiary quantity} * \frac{\text{numerator}_{\text{tertiary quantity}}}{\text{denominator}_{\text{tertiary quantity}}}$$



If no base quantity is specified, a conversion cannot be performed.

Usage with data collection and posting

Data collection can generally be performed using all 4 quantity units. But to identify the quantities, the system does not use the unit, but the acronym entered in the terminal dialog. With automatic quantity recording (e.g. data collection via CT-UMPS), data is always recorded in the primary quantity unit.

If exactly one quantity unit is used for data collection, data is first converted into the base quantity unit and from the base quantity unit into the other quantity units.

If data is recorded in more than one quantity unit, the first unit is converted based on the following sequence:

- Base quantity unit
- Primary input quantity unit
- Secondary input quantity unit
- Tertiary input quantity unit

Additionally recorded quantities in other quantity units are not recalculated but saved as recorded.

If parallel quantity units are kept, but no conversion factors are specified, then these quantity units are calculated based on the material-specific conversion rules. Data is converted using formulas generally referring to material properties (contents of user fields, such as the width or the relevant fields included in the backlog of orders, e.g. partitioning).

Target quantity update: If you transfer the actual quantity to the subsequent OP, the system first converts the quantity into the base quantity unit within the forwarding OP, then this quantity is transferred to the subsequent operation and here again reconverted into the primary input quantity unit or the alternative quantity units.



A quantity 0 is generally not converted into alternative quantity units even if a value that is not 0 could be calculated (e.g. by means of a formula).

Usage in MPL

Alternative quantity units can be recorded with coil-based manufacturing, if you make a respective configuration for the operation.

Example:

- Base quantity unit m2
- Lengths are recorded as primary quantity unit
- Secondary quantity unit weight
- Tertiary quantity unit piece

Scrap quantities are managed in these 4 quantity units.

Example 1

You want to manage the quantities of an operation in the following quantity units:

- Base quantity unit: [M2] square meter
- Primary quantity unit: [PCS] piece

With 1 [PCS] = 2 [M2].

Conversion factors for base quantity

Primary quantity, numerator	2	Primary quantity, denominator	1
Secondary quantity, numerator	0	Secondary quantity, denominator	0
Tertiary quantity, numerator	0	Tertiary quantity, denominator	0

Example 2

You want to manage the quantities of an operation in the following quantity units:

- Base quantity unit: [CM] centimeter
- Primary quantity unit: [PCS] piece
- Secondary quantity unit: [KAR] box/carton
- Tertiary quantity unit: [PAL] pallet

With:

- 1 [PCS] = 0,5 [CM]
- 1 [KAR] = 100 [PCS] = 50 [CM]
- 1 [PAL] = 20 [KAR] = 2000 [PCS] = 1000 [CM]

Conversion factors for base quantity

Primary quantity, numerator	1	Primary quantity, denominator	2
Secondary quantity, numerator	50	Secondary quantity, denominator	1
Tertiary quantity, numerator	1000	Tertiary quantity, denominator	1

Example 3

You want to manage the quantities of an operation in the following quantity units:

- Base quantity unit: [M2] square meter
- Primary quantity unit: [M] meter
- Secondary quantity unit: [KG] kilogram
- Tertiary quantity unit: [PCS] piece

With:

- 1 [M] = 2,5 [M2]
- 1 [KG] = 4,5 [M] = 11,25 [M2]
- 1 [PCS] = 0,5 [KG] = 2,25 [M] = 5,625 [M2]

Conversion factors for base quantity

Primary quantity, numerator	5	Primary quantity, denominator	2
Secondary quantity, numerator	45	Secondary quantity, denominator	4
Tertiary quantity, numerator	45	Tertiary quantity, denominator	8

Example 4

You want to manage the quantities of an operation in the following quantity units:

- Primary quantity unit: [ST] piece
- Secondary quantity unit: [KAR] box/carton

With:

- 1 [KAR] = 240 [ST]



The system always uses the base quantity unit for the conversion from one quantity unit (e.g. primary quantity unit) into another quantity unit (e.g. secondary quantity unit). For this reason, you **MUST** store a unit in the order header.

Conversion factors for base quantity

Primary quantity, numerator	1	Primary quantity, denominator	1
Secondary quantity, numerator	240	Secondary quantity, denominator	1
Tertiary quantity, numerator	0	Tertiary quantity, denominator	0